

10.1 DEMONSTRATIVE GEOMETRY

(i) Define Demonstrative Geometry

'Demonstrative Geometry' is a branch of Mathematics in which statements concerning geometrical figures (called theorems) are proved through logical Reasoning.

The word 'Demonstrate' comes from the Latin word 'Demonstratum' which means 'to prove with certainty'.

10.1.1 Reasoning

(ii) Describe the Basics of reasoning

Reasoning is the process of thinking about something in a logical way in order to form a conclusion or judgement. Reasoning may be Inductive or Deductive.

Quite often we come across situations where we draw conclusions based on observations, e.g., we visited a new country and noticed that each school bus we came across was of yellow colour. Upon our return we gave a concluding statement:

'All school busses in that country are of yellow colour.'

Such a reasoning, in which we generalize specific observations into universal statements, is called Inductive Reasoning.

The problem with inductive reasoning is that, even in the best (or strongest) cases the truth of the statement is not guaranteed; as in the above example there could have been some rare red buses which we couldn't have spotted in the whole trip.

Now consider the following statement:

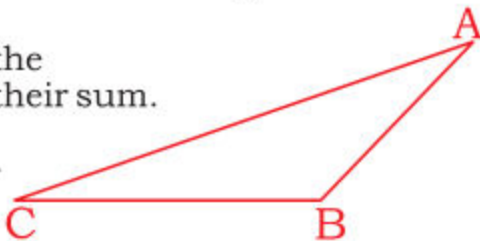
'Sum of the measures of three angles of a triangle is 180° .'

In order to prove this statement we can draw triangles of different kinds and different sizes. Measure their three angles. Then find their respective sum.

Activity: Let each student measure the three angles of this triangle and find their sum.

$m\angle A =$ _____, $m\angle B =$ _____

$m\angle C =$ _____, Sum = _____



We see that some students got the sum 179° . Others got the sum 181° . Only a few students got the sum 180° . Even one or two students got sum as 178° or 182° .

So from this activity we come to know that **Inductive Reasoning** is often not very reliable and we have to follow many precautionary measures to use this kind of reasoning. It is also time consuming.

Contrary to this, in **Deductive Reasoning** we deduce particular results from general ones. Generally we start from a set of statements already believed to be true and construct a series of deductions and use to establish proof for new statements.

We study Geometry through Deductive Reasoning because the new statements can be deduced with certainty.

The whole framework of deductive reasoning is based on the following four elements known as the *Basics of Reasoning*.

1. Undefined Terms: Some geometrical terms are accepted without definition. These are called **Undefined Terms**.

Point, Line, Plane and Space are **Undefined Terms**.

2. Defined Terms: With the help of undefined terms, we define various new geometrical terms. For example: Line Segment, Ray, Angle, Triangle, Perpendicular, Bisector etc. are defined terms. Many of these terms have been defined in earlier classes.

3. Assumptions (or Fundamental Agreements):

Some mathematical statements are accepted without proof. These are called Assumptions or **Fundamental Agreements**. For example: 'Every number is equal to itself' Or 'Two points determine a unique line.' In section 10.1.2, we will introduce two types of assumptions and will enumerate all the assumptions that we shall use for reasoning in this unit.

4. Propositions: With the help of *Undefined Terms*, *Defined Terms*, and *Fundamental Agreements*, we can develop new statements. These statements are termed as **propositions** and may be proved true or false through logical reasoning. A generic proposition about some geometrical concept, if proved true, is called a theorem.

10.1.2 Axioms, Postulates and Theorem

(iii) **Describe the Types of Assumptions (Axioms and Postulates):**

Assumptions or Fundamental Agreements are of two kinds:

A. Axioms: Fundamental Agreements which are related to

numbers are called Axioms. Following are the axioms which will be used in this unit later on.

Axiom 1. Every number is equal to itself (Reflexive Property of Equations) i.e.,

$$\begin{array}{l} x = x \quad \forall x \in \mathbb{R} \\ \text{or} \quad \angle A \cong \angle A \quad (\text{Identity Congruence}) \end{array}$$

Axiom 2. Symmetric Property of Equations. i.e.,

$$x = y \Leftrightarrow y = x \quad \forall x, y \in \mathbb{R}$$

Axiom 3. Transitive Property of Equations. i.e.,

$$x = y \text{ and } y = z \Rightarrow x = z \quad \forall x, y, z \in \mathbb{R}$$

Axiom 4. Equals added to equals remain equal (Addition Property of Equations) i.e.,

$$x = y \Rightarrow x + z = y + z \quad \forall x, y, z \in \mathbb{R}$$

Axiom 5. Equals subtracted from equals remain equal (Subtraction Property of Equations) i.e.,

$$x = y \Rightarrow x - z = y - z \quad \forall x, y, z \in \mathbb{R}$$

Axiom 6. If Equals are multiplied by equals, their products are equal (Multiplication Property of Equations) i.e.,

$$x = y \Rightarrow xz = yz \quad \forall x, y, z \in \mathbb{R}$$

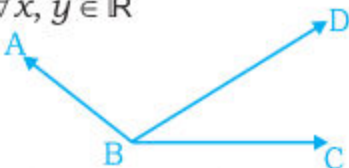
Axiom 7. If equals are divided by a non-zero number, their quotients are equal i.e.,

$$x = y \text{ and } z \neq 0 \Rightarrow \frac{x}{z} = \frac{y}{z} \quad \forall x, y \in \mathbb{R}$$

Axiom 8. Whole is greater than its part. i.e.,

$$m\angle ABC = m\angle ABD + m\angle CBD$$

$$\Rightarrow m\angle ABC > m\angle ABD, m\angle ABC > m\angle CBD$$



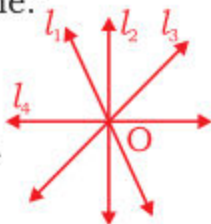
B. Postulates: Fundamental Agreements which are related to geometrical figures are called Postulates.

Postulate 1. One and only one line passes through two distinct points. It is also stated as: 'Two points determine a line.'



Postulate 2. Countless lines can pass through a point.

Lines $l_1, l_2, l_3, \dots, l_n$ are passing through a point O.



Postulate 3. One and only plane passes through three points not in a line.

Postulate 4. If two points of a line lie on a plane, the whole line lies on the plane.

[This postulate tells us that surface of plane is smooth and it is infinite on all sides.]

Postulate 5 (Distance Postulate). If A and B are any two points of a plane, then distance between them will be:

- (i) 0 (zero), if A lies on B, i.e., $A = B$ [read as A coincides with B]
- (ii) a 'Positive' real number when A and B are distinct points.

The distance from point A to point B is denoted as $|\overline{AB}|$ or $m\overline{AB}$, m stands for 'measure'. It may be noted that $m\overline{AB} = m\overline{BA}$ or $|\overline{AB}| = |\overline{BA}|$.

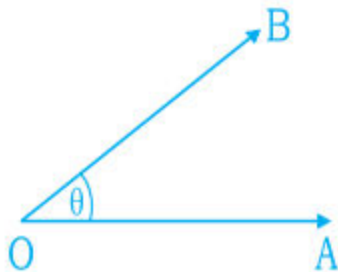
Postulate 6. A line segment can be bisected at one and only one point.



[This postulate tells that on a line segment AB, there is only one point (say P) between A and B such that $m\overline{AP} = m\overline{BP}$]

Postulate 7 (Angle Construction Postulate).

If one arm of an angle is given, one and only one ray can be drawn on one side of it making an angle of given measure θ between 0° and 180° , i.e., $0^\circ < \theta < 180^\circ$.

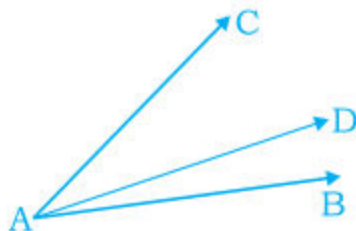


Postulate 8 (Angle Addition Postulate).

The sum of the measures of two adjacent angles is equal to the measure of the angle formed by their non-common arms.

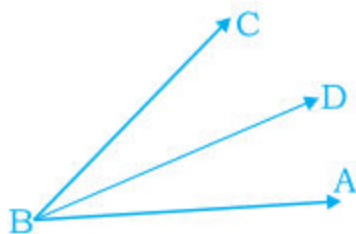
In this figure, $\angle BAD$ and $\angle CAD$ are two adjacent angles. So,

$$m\angle BAD + m\angle CAD = m\angle BAC \text{ (or } m\angle CAB).$$



Postulate 9 (Unique Bisector of an angle Postulate). One and only one bisector of an angle can be drawn. In this figure, $\overline{BD}$ is the bisector of $\angle ABC$.

$$\text{So, } m\angle ABD = m\angle CBD = \frac{1}{2} m\angle ABC$$



Postulate 10. At a given point in a line, one and only one perpendicular can be drawn to the line.

Postulate 11. Geometrical figures which can be made to coincide with one another are congruent.

Postulate 12. A geometrical figure may be moved from one position to another without any change in shape and size.

Postulate 13 (Playfair's Axiom). Through a point not on a line, one and only one line can be drawn parallel to the line. It is also stated as: 'Two intersecting lines cannot be parallel to a third line.'

(iv) Describe parts of a proposition

When a statement about any geometrical subject is proposed for discussion, it is called a **proposition**. Propositions are either **Theorems** or **Problems** (also called **Riders**).

A Proposition (Theorem or Rider) consists of two parts:

- (i.) **Hypothesis** or that which is assumed to be true
- (ii.) **Conclusion** or that which is to be proved.

(v) Describe the meanings of geometrical **theorem**, **corollary** and **converse of a theorem**.

(a) Theorem

A Theorem is a proposition in which some geometrical fact is stated. The general statement of a theorem is called Enunciation, e.g.,

'If two lines intersect, then the opposite vertical angles are congruent.'

In this proposition (i.e., theorem) **Hypothesis** is: "If two lines intersect" and **Conclusion** is: "then the opposite vertical angles are congruent." Usually, the **hypothesis** starts with 'If' and **Conclusion** with 'then'; but sometimes both **hypothesis** and **Conclusion** are mixed up in a single simple sentence, e.g.,

'Sum of the measures of the three angles of a triangle is 180° .'

Here the hypothesis is: 'If the angles are the three angles of a triangle' and **conclusion** is: 'then the sum of the measures of these angles is 180° .'

(b) Corollary

A Corollary (abbreviated as Cor.) is a fact which readily follows from a proved theorem. Such propositions are so trivial that we do not label them as separate theorems. These are the deductions made from a proposition or a theorem.

(c) Converse of a Theorem

One theorem is said to be the **Converse** of the other when the **hypothesis** of each is the **conclusion** of the other. In this text book, Theorems 1 and 2 are converse of one another. Also Theorems 7 and 8 are converse of one another.

In order **to prove a Theorem**, a Corollary or a Rider (given as an exercise) we follow certain fixed steps which are as under:

- 1. Figure or Diagram:** According to the Enunciation, a neat figure is drawn showing the requirements of the Hypothesis as well as that of the Conclusion.
- 2. Given or Data:** This is the particular enunciation of the hypothesis and is interpreted in terms of the figure in Step 1 above. It tells us what is the hypothesis in particular.
- 3. Required to Prove:** Here the Conclusion is written in particular terms according to the figure drawn in Step 1 above.
- 4. Construction:** Sometimes we need certain addition in the figure that helps us in proving the theorem. So we do certain construction.
- 5. Proof:** This is the last part of a proof of a theorem (or a Problem) in which the given proposition is proved logically with the help of definitions, postulates, axioms, and the given data, or theorems proved earlier. We write statements one after other in one column and give reasons in another column opposite to it.
- 6.** At the end of a Proof, we usually write **Q.E.D.**, the abbreviation of **Quod Erat Demonstrandum** meaning '**which was to be proved**'.

EXERCISE 10.1

1. Fill in the blanks with appropriate words:

- (a) Demonstrative Geometry is a _____ of Mathematics in which _____ on geometry are proved through _____.
- (b) Reasoning may be _____ or _____.
- (c) In _____ reasoning, we draw conclusions based on observations.
- (d) In _____ reasoning, we deduce particular results from general results and use _____ to prove new statements.
- (e) Assumptions are those statements which are accepted _____.
- (f) Fundamental Agreements are of two kinds. 1. _____
2. _____.
- (g) Fundamental Agreements which are related to geometrical figures are called _____.
- (h) The four basics of Reasoning are:
1. _____ 2. _____ 3. _____ 4. _____

2. Write 'True' or 'False' against each statement:

- (a) One and only straight line can be drawn through three distinct points.
- (b) If P and Q are coincident points, symbolically we write $P = Q$.
- (c) Two points do not determine a line.
- (d) Postulate 4 tells us that the surface of the plane is smooth and it is infinite on all sides.
- (e) Complement of an angle of x° is $(90 - x)^\circ$.
- (f) 'If two quantities are equal to the same quantity, they are equal to one another' is the same as Transitive Property of Equations.

3. Name the two parts of a Theorem or Proposition.

10.2 THEOREMS

Prove the following theorems along with corollaries and apply them to solve appropriate problems.

Here proofs of prescribed theorems along with corollaries and their applications to solve appropriate problems are given:*

THEOREM 1

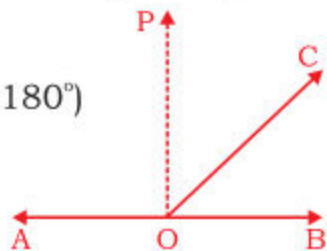
If a straight line stands on another straight line, the sum of measures of two angles so formed is equal to two right angles.

Given: $\overrightarrow{OC}$ stands on $\overleftrightarrow{AB}$

To prove: $m\angle AOC + m\angle BOC = 2\text{rt}\angle\text{s (or } 180^\circ\text{)}$

Construction: Through O, Draw $\overrightarrow{OP} \perp \overleftrightarrow{AB}$

i.e., $m\angle AOP = 1\text{rt}\angle = m\angle BOP$

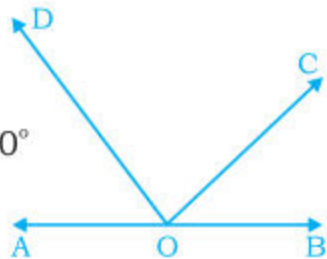


Proof:	Statements	Reasons
1.	$m\angle AOC = m\angle AOP + m\angle POC$	i. Angle Addition Postulate
2.	$m\angle BOC = m\angle BOP - m\angle POC$	ii. $m\angle BOC + m\angle POC = m\angle POB$ (AAP)
3.	$m\angle AOC + m\angle BOC = m\angle AOP + m\angle BOP$ $= 1\text{rt}\angle + 1\text{rt}\angle$ $= 2\text{rt}\angle\text{s (or } 180^\circ\text{)}$	iii. Adding (1) and (2), $m\angle POC$ cancels Each is a right angle
Q.E.D.		

Cor. 1. The sum of measures of all the angles at a point in a given straight line on the same side of it is equal to two right angles.

Given: $\overrightarrow{OC}$ and $\overrightarrow{OD}$ stand on $\overleftrightarrow{AB}$ on the same side of it making angles $\angle AOD, \angle DOC, \angle BOC$

To Prove: $m\angle AOD + m\angle DOC + m\angle BOC = 180^\circ$



Note: Statements No. 1,2,3, etc have the reasons i, ii, iii, etc respectively, in each and every theorem.

Proof:

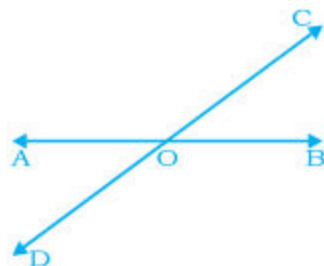
Statements	Reasons
1. $m\angle AOC + m\angle BOC = 180^\circ$	i. By Theorem 1
2. But $m\angle AOC = m\angle AOD + m\angle DOC$	ii. Angle Addition Postulate
3. $m\angle AOD + m\angle DOC + m\angle BOC = 180^\circ$	iii. Putting the value of $m\angle AOC$ in (1)
Q.E.D	

Cor. 2. If two lines intersect, the sum of the measures of the four angles is equal to four right angles.

Given: $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ intersect at O.

To Prove :

$$m\angle AOC + m\angle BOC + m\angle BOD + m\angle AOD = 360^\circ$$

Proof:


Statements	Reasons
1. $m\angle AOC + m\angle BOC = 180^\circ$	i. By Theorem 1
2. $m\angle BOD + m\angle AOD = 180^\circ$	ii. By Theorem 1
3. $m\angle AOC + m\angle BOC + m\angle BOD + m\angle AOD = 360^\circ$	iii. Adding (1) and (2)
Q.E.D.	

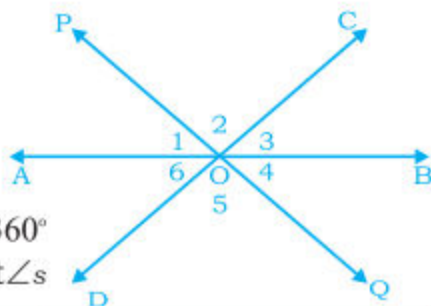
Cor. 3. If a number of straight lines meet at a point, the sum of measures of all the angles between successive lines is equal to four right angles.

Given: $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$ and $\overleftrightarrow{PQ}$ intersect at point O.

To Prove :

$$m\angle 1 + m\angle 2 + m\angle 3 + m\angle 4 + m\angle 5 + m\angle 6 = 360^\circ$$

$$= 4\text{rt}\angle s$$

Proof:


Statements	Reasons
1. $m\angle 1 + m\angle 2 + m\angle 3 = 180^\circ$	i. By Theorem 1 Cor. 1
2. $m\angle 4 + m\angle 5 + m\angle 6 = 180^\circ$	ii. By Theorem 1 Cor. 1
3. $m\angle 1 + m\angle 2 + m\angle 3 + m\angle 4 + m\angle 5 + m\angle 6 = 360^\circ = 4\text{rt}\angle s$	iii. Adding (1) and (2)
Q.E.D.	

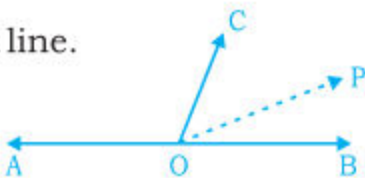
THEOREM 2

If the sum of measures of two adjacent angles is equal to two right angles, the external arms of the angles are in a straight line.

Given: Two adjacent angles $\angle AOC$ and $\angle BOC$
such that $m\angle AOC + m\angle BOC = 180^\circ = 2rt\angle s$

To prove: $\overrightarrow{OA}$, $\overrightarrow{OB}$ are in the same straight line.

Construction: If $\overrightarrow{OB}$ is not in the same straight line with $\overrightarrow{OA}$, produce $\overrightarrow{AO}$ to point P.



Proof:

Statements	Reasons
1. AOP is a straight line and OC stands on it	i. Construction, Supposition
2. $m\angle AOC + m\angle POC = 180^\circ$	ii. By Theorem 1
3. But $m\angle AOC + m\angle BOC = 180^\circ$	iii. Hypothesis
4. $m\angle AOC + m\angle POC = m\angle AOC + m\angle BOC$	iv. Axiom 3
5. Or $m\angle POC = m\angle BOC$	v. Canceling $m\angle AOC$ from both sides
6. Which is impossible unless OP and OB coincide Hence OA, OB are in the same straight line.	vi. Our Supposition in the construction is wrong. So, given facts are always true.

Q.E.D.

Note: Method used in the above proof is called '**Reduction ad Absurdum**' Method meaning 'Reduction to Absurdity' Method.

EXERCISE 10.2

1. In the figure of Theorem 1.

- Name the angle which is supplement of $\angle BOC$
- Name the angle which is complement of $\angle BOC$
- If $m\angle BOC = 50^\circ$, find measures of its complement and supplement.

2. In the figure of Theorem 1, Cor. 1, let $m\angle BOC = 45^\circ$, $m\angle AOD = 2x$, $m\angle COD = 3x$. Find the measures of $\angle AOC$, $\angle COD$.

[**Hint:** Solve for x the Eq: $2x + 3x + 45 = 180$].

3. In the figure of Theorem 1 Cor. 2.

(a) Let $m\angle BOC = x = 40^\circ$, find the remaining angles

[Hint: Let $m\angle AOC = y$, $m\angle AOD = z$, $m\angle BOD = w$,

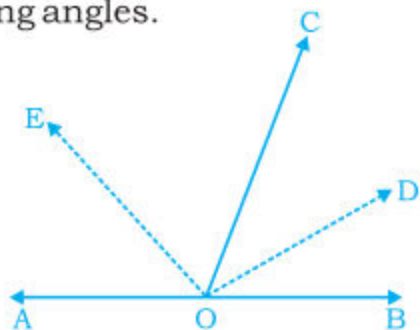
Now solve $w + 40^\circ = 180^\circ$, $y + 40^\circ = 180^\circ$, then $z + 140 = 180$ (why?)]

(b) Let $m\angle BOC = x = 45^\circ$, find the remaining angles.

4. Bisectors of two adjacent supplementary angles are perpendicular to each other.

Given: $\overrightarrow{OD}$, $\overrightarrow{OE}$ are bisectors of two adjacent angles AOC, BOC.

To Prove: $\overrightarrow{OD} \perp \overrightarrow{OE}$



Proof:

Statements	Reasons
1. $m\angle AOC + m\angle BOC = 180^\circ$	i. Given (Two adjacent suppl. $\angle$ s)
2. $\frac{1}{2} m\angle AOC + \frac{1}{2} m\angle BOC = \frac{1}{2} (180^\circ)$	ii. Multiplying both sides of Eq. by $\frac{1}{2}$
3. or $m\angle COE + m\angle COD = 90^\circ$	iii. $\frac{1}{2} m\angle AOC = m\angle COE$ and $\frac{1}{2} m\angle BOC = m\angle COD$
4. or $m\angle EOD = 90^\circ$	iv. Angle Addition Postulate
5. or $OD \perp OE$	v. Two rays are perpendicular, If they form a rt $\angle$
Q.E.D.	

5. If the bisectors of two adjacent angles are perpendicular to each other, the angles are supplementary angles. (Converse of problem 4 above).

[**Hint:** Since it is converse of problem 4, if we start from bottom of solution of problem 4 and end at the top with slight change somewhere for example you have to multiply by 2 at step 3.]

6. In the figure of problem 4, name all the pairs of complementary angles.

THEOREM 3

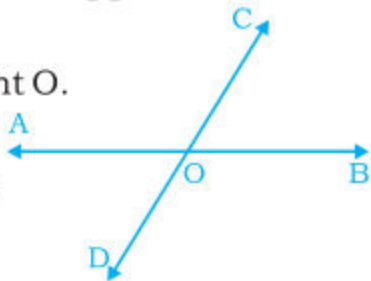
If two lines intersect each other, then the opposite vertical angles are congruent.

Given: Two lines AB and CD intersect at point O.

To Prove:

- (i) $\angle BOC \cong \angle AOD$ (ii) $\angle AOC \cong \angle BOD$

Proof:



Statements	Reasons
1. $m\angle AOC + m\angle BOC = 180^\circ$	i. By Theorem 1
2. $m\angle AOD + m\angle AOC = 180^\circ$	ii. By Theorem 1
3. $\therefore m\angle AOC + m\angle BOC = m\angle AOD + m\angle AOC$	iii. Transitive Prop. of Eq.
4. $\therefore m\angle BOC = m\angle AOD$	iv. Cancelling $m\angle AOC$ from both sides
5. or $\angle BOC \cong \angle AOD$	v. If two angles are equal in measure, they are congruent.
6. Similarly, we can prove that $\angle AOC \cong \angle BOD$	vi. By the above process.
Q.E.D.	

EXERCISE 10.3

1. In the figure of Theorem 3, if $m\angle BOC = 70^\circ$, find the measures of other angles.

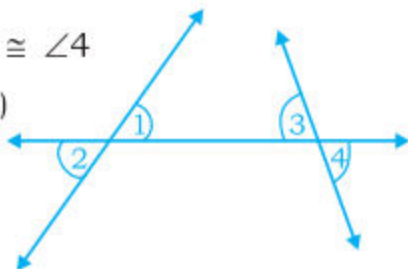
[Comment: In the previous exercise we used Theorem 1 to find other angles, because until then we had not learnt Theorem 3. But now we can use Theorem 3 together with Theorem 1]

Solution: $m\angle AOD = m\angle BOC = 70^\circ$ (By Theorem 3) and $\angle AOC$ is the supplement of $\angle BOC$. Therefore, $m\angle AOC = 180^\circ - 70^\circ = 110^\circ$ (By Theorem 1) and $m\angle BOD = 110^\circ$ (By Theorem 3).

2. Draw two lines intersecting one another at an angle of 30° , find the measures remaining angles as discussed above.

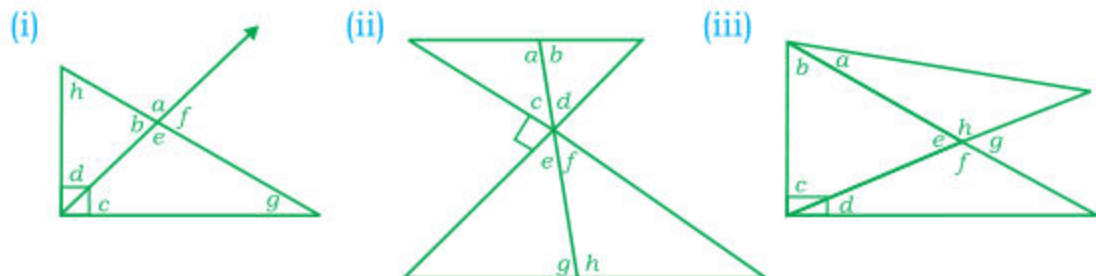
3. In this figure $\angle 1 \cong \angle 3$, Prove that $\angle 2 \cong \angle 4$

[**Hint:** First prove $\angle 2 \cong \angle 1$ and $\angle 3 \cong \angle 4$ (Th. 1) then use Axioms 3 since $\angle 1 \cong \angle 3$ is given.]

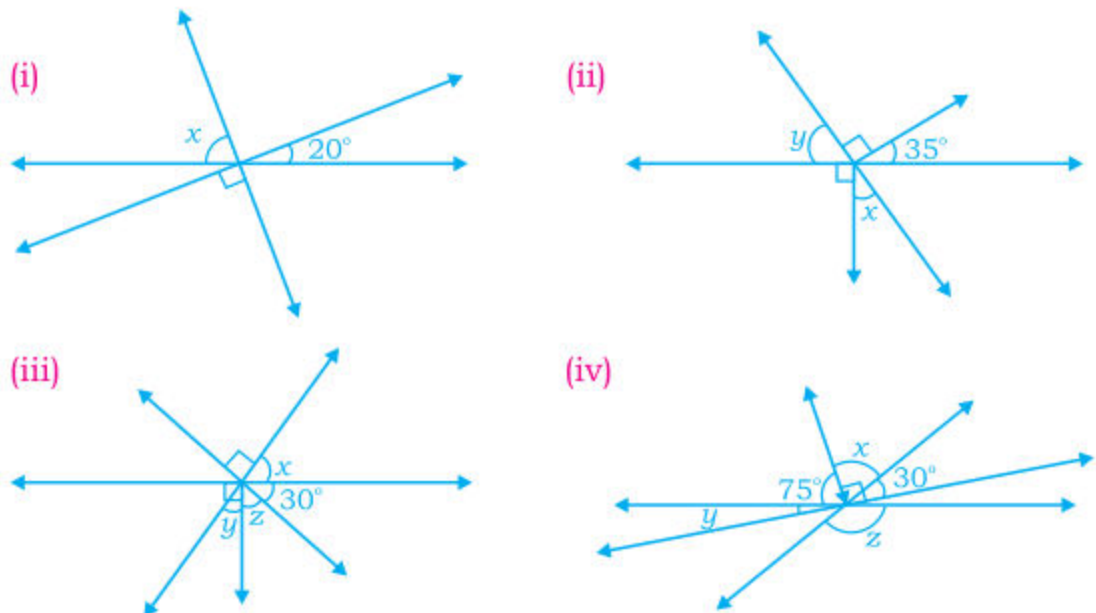


4. If the bisectors of a pair of congruent angles having the same vertex are two opposite rays, then the sides of the angles are two intersecting lines.

5. In the following figures, name the complementary angles, the supplementary angles, and the vertically opposite angles:



6. Find the degree measure for the angles marked x , y and z .

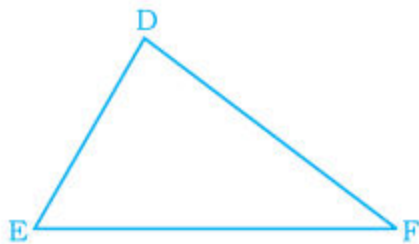
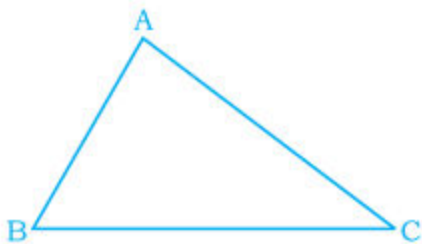


Definitions:**One-One Correspondence between two triangles.**

Every triangle has three vertices and three sides, so it is always possible to establish One-One Correspondence between the sides, angles and vertices of two triangles.

Symbol ' $\leftrightarrow$ ' is used for One-One (or 1 - 1) Correspondence.

The statement ' $\triangle ABC \leftrightarrow \triangle DEF$ ' means that: A corresponds to D, B to E, C to F, $\overline{AB}$ corresponds to $\overline{DE}$, $\overline{BC}$ to $\overline{EF}$, and $\overline{AC}$ to $\overline{DF}$.



' $\triangle ABC \leftrightarrow \triangle EDF$ ' is not the same as ' $\triangle ABC \leftrightarrow \triangle DEF$ ' because here A corresponds to E, B to D, and so on.

Congruence of Triangles:

Two Triangles are said to be congruent if their corresponding angles and sides are congruent. Conversely, if two triangles are congruent, then their corresponding angles and sides are congruent. Symbol used for congruence is ' $\cong$ '.

Note 1. If one correspondence of two triangles is a congruence, it is not necessary that other correspondences between them are also congruences.

Note 2. A triangle is congruent to itself. This is called **Identity Congruence** of triangles.

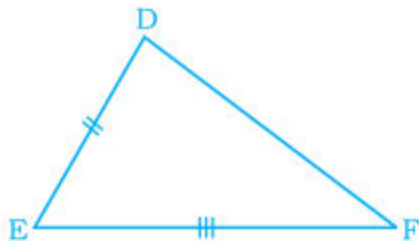
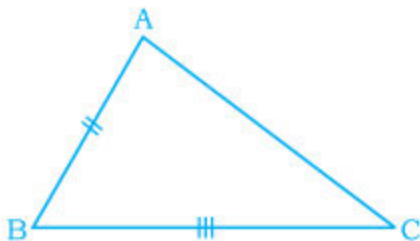
For example: $\triangle ABC \cong \triangle ABC$

Note 3. $\triangle ABC \cong \triangle DEF$ implies $\triangle DEF \cong \triangle ABC$ (**Symmetric Property of Congruence**).

Note 4. If $\triangle ABC \cong \triangle DEF$ and $\triangle DEF \cong \triangle PQR$, then $\triangle ABC \cong \triangle PQR$ (**Transitive Property of Congruence**).

THEOREM 4

In any correspondence of two triangles, if two sides and included angle of one triangle are congruent to the corresponding sides and included angles of the other, the two triangles are congruent.



Given: In $\triangle ABC \leftrightarrow \triangle DEF$

(i) $\overline{AB} \cong \overline{DE}$

(ii) $\angle B \cong \angle E$

(iii) $\overline{BC} \cong \overline{EF}$

To Prove: $\triangle ABC \cong \triangle DEF$

Proof:

Statements	Reasons
1. Apply $\triangle ABC$ on $\triangle DEF$ so that Point B falls on point E and BC along EF	i. Postulate 12
2. $BC \cong EF$	ii. Given
3. $\therefore$ Point C coincides with point F	iii. By definition of Congruence
4. $\angle B \cong \angle E$	iv. Given
5. $\therefore$ BA falls along ED	v. By Angle Construction Postulate
6. $AB \cong DE$	vi. Given
7. $\therefore$ A must coincide with point D	vii. By definition of Congruence
8. $\triangle ABC$ coincides with $\triangle DEF$	viii. As proved above
9. Hence, $\triangle ABC \cong \triangle DEF$.	ix. By Postulate 11
Q.E.D	

Note: This theorem is also quoted as 'S.A.S. $\cong$ S.A.S.'

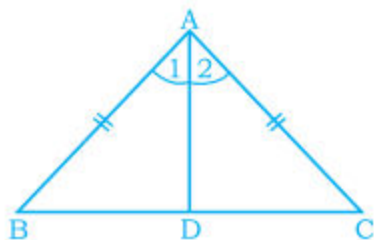
THEOREM 5

If two sides of a triangle are congruent, then the angles opposite to these sides are congruent.

Given: In $\triangle ABC$, $\overline{AB} \cong \overline{AC}$

To Prove : $\angle B \cong \angle C$

Construction: Draw $\overline{AD}$ the bisector of $\angle A$ meeting $\overline{BC}$ at point D.



Proof:

Statements	Reasons
1. In $\triangle ABD \leftrightarrow \triangle ACD$	i. 1-1 correspondence between two $\triangle$ s
(A) $AB \cong AC$	(a) Given
(B) $\angle 1 \cong \angle 2$	(b) Construction
(C) $AD \cong AD$	(c) Common (Identity Congruence)
2. $\triangle ABD \cong \triangle ACD$	ii. S.A.S. $\cong$ S.A.S.
3. $\therefore \angle B \cong \angle C$	iii. By the congruence of triangle
Q.E.D.	

Note: Theorem 5 can also be stated as:

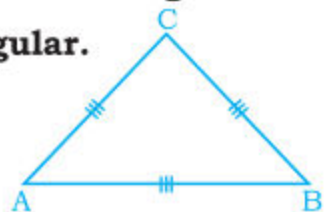
'In an Isosceles triangle, angles at the base are congruent.'

Cor. 1. An equilateral triangle is also equiangular.

Given: $\triangle ABC$ in which $\overline{AB} \cong \overline{BC} \cong \overline{AC}$

To Prove : $\angle A \cong \angle B \cong \angle C$

Proof:



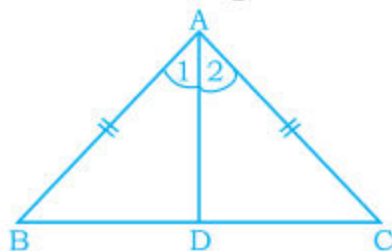
Statements	Reasons
1. In $\triangle ABC$, $BC \cong AC$	i. Given
2. $\therefore \angle A \cong \angle B$	ii. By Theorem 5
3. Similarly, $\angle B \cong \angle C$	iii. $AB \cong AC$
4. $\therefore \angle A \cong \angle B \cong \angle C$	iv. Combining (2) and (3)
Q.E.D.	

Cor. 2. In an Isosceles triangle, the bisector of the angle at the vertex is the right bisector of the base.

Given: In $\triangle ABC$, $\overline{AB} \cong \overline{AC}$, and
 $\overline{AD}$ is the bisector of $\angle A$ meeting
 $\overline{BC}$ at D.

To Prove: $\overline{BD} \cong \overline{DC}$, and $\overline{AD} \perp \overline{BC}$

Proof:



Statements	Reasons
1. In $\triangle ABD \leftrightarrow \triangle ACD$	i. 1-1 correspondence between two $\triangle$ s.
(A) $AB \cong AC$	(a) Given
(B) $\angle 1 \cong \angle 2$	(b) Given
(C) $AD \cong AD$	(c) Common
2. $\therefore \triangle ABD \cong \triangle ACD$	ii. By Theorem 4 (S.A.S. $\cong$ S.A.S.)
3. $\therefore BD \cong DC$ and $\angle ADB \cong \angle ADC$	iii. By the congruence of triangles
4. But $m\angle ADB + m\angle ADC = 180^\circ$	iv. By Theorem 1
5. $\therefore m\angle ADB = 90^\circ = m\angle ADC$	v. By Definition of rt. angle
6. $\therefore AD \perp BC$	vi. By Definition of Perpendicular
7. $\therefore AD$ is Perp. Bisector of BC	vii. AD bisects BC at right angles
Q.E.D.	

THEOREM 6

An exterior angle of a triangle is greater in measure than either of its opposite interior angles.

Given: $\triangle ABC$ with exterior angle ACD

To Prove:

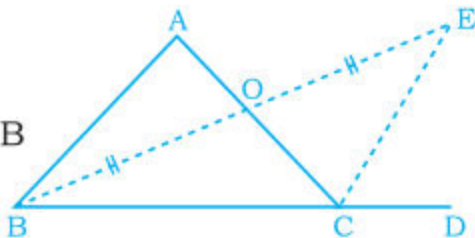
$$m\angle ACD > m\angle A, m\angle ACD > m\angle B$$

Construction:

Let point O be the mid-point of $\overline{AC}$.

Draw $\overline{BO}$ and produce it to point E. So that, $m\overline{BO} = m\overline{OE}$.

Join point C to point E.



Proof:

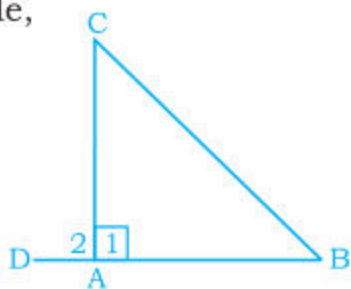
Statements	Reasons
1. In $\triangle AOB \leftrightarrow \triangle COE$	i. 1-1 correspondence between two $\triangle$ s.
(A) $BO \cong OE$	(a) Construction
(B) $\angle AOB \cong \angle COE$	(b) Vertical angles
(C) $AO \cong CO$	(c) Construction
2. $\therefore \triangle AOB \cong \triangle COE$	ii. S.A.S. $\cong$ S.A.S.
3. $\therefore m\angle A = m\angle ACE$	iii. By the congruence of triangles
4. But $m\angle ACD = m\angle ACE + m\angle ECD$	iv. Angle addition Postulate
5. $\therefore m\angle ACD > m\angle ACE$	v. Whole is greater than its part
6. $\therefore m\angle ACD > m\angle A$	vi. $m\angle A = m\angle ACE$ (Proved above)
7. Similarly, $m\angle ACD > m\angle B$	
Q.E.D.	

Cor. 1. In a triangle, if one angle is a right angle, then the other angles are acute angles.

Given: $\triangle ABC$ is a right-triangle at Point A.

To Prove : $m\angle B < 90^\circ$ and $m\angle C < 90^\circ$

Construction: Produce BA to any point D.

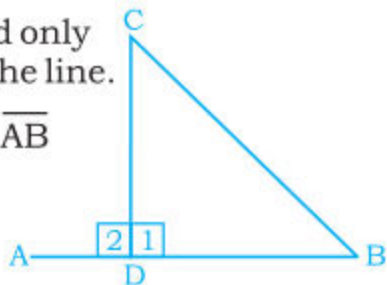
**Proof:**

Statements	Reasons
1. $m\angle 2 = 90^\circ$	i. $m\angle 1 + m\angle 2 = 180^\circ$ and $m\angle 1 = 90^\circ$
2. But $\angle 2$ is exterior angle of $\triangle ABC$	ii. By Definition
3. $\therefore m\angle B < m\angle 2 \Rightarrow m\angle B < 90^\circ$	iii. By Theorem 6
4. Similarly, $m\angle C < 90^\circ$	iv. By the above reason
Q.E.D.	

Cor. 2. From a point not on the line, one and only one perpendicular can be drawn to the line.

Given: C is any point outside $\overline{AB}$, and $\overline{CD} \perp \overline{AB}$

To Prove: CD is the only perpendicular.



Proof:

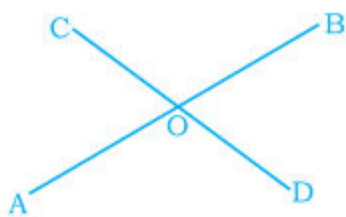
Statements	Reasons
1. If CD is not the only perpendicular. Suppose CB is also perpendicular on AB	i. Supposition
2. But in $\triangle DBC$, $m \angle B < \angle 2$	ii. By Theorem 6
3. $\Rightarrow m \angle B < 90^\circ$	iii. $m \angle 2 = 90^\circ$
4. $\therefore$ CB is not perpendicular on AB	iv. Our supposition is wrong
5. Hence CD is the only perpendicular on AB	v. As proved above.
Q.E.D	

EXERCISE 10.4

1. In the adjacent figure, O is the mid point of line segments AB and CD.

Prove that $\overline{AC} \cong \overline{BD}$

[**Hint:** Join point A to point C and point B to point D. Prove $\triangle AOC \cong \triangle BOD$ by S.A.S. $\cong$ S.A.S.]



2. Two line segments bisect one another. Prove that the line segments joining their end-points are congruent. [This is the general statement of problem 1 above]

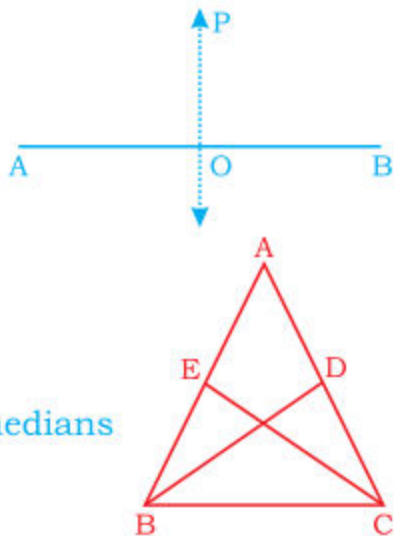
3. Prove that the diagonals of a rectangle are congruent.

[**Hint:** Let $\overline{AC}$ and $\overline{BD}$ be the diagonals of rectangle ABCD. Prove that $\triangle ABC \cong \triangle BAD$ by S.A.S. Theorem]

4. In an Isosceles triangle, the bisector of the angle at the vertex angle is the right bisector of the base. Prove.

5. Every point on the perpendicular bisector of a line segment is equidistant from the end-points of the line segment.

[Here $\overleftrightarrow{OP}$ is the perpendicular bisector of $\overline{AB}$. We have to prove that $\overline{AP} \cong \overline{BP}$. So join $\overline{AP}$ and $\overline{BP}$, then prove that triangles AOP and BOP are congruent by S.A.S. Theorem]



6. Medians to the congruent sides of an isosceles triangle are congruent.

Given: In $\triangle ABC$, $\overline{AB} \cong \overline{AC}$ and $\overline{BD}$, $\overline{CE}$ are medians

To Prove That: $\overline{BD} \cong \overline{CE}$

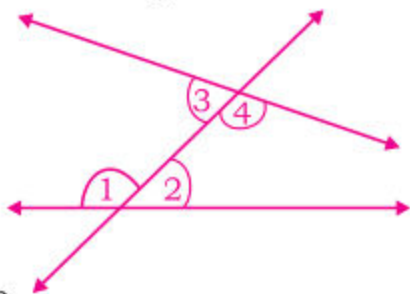
Proof:

Statements	Reasons
1. In $\triangle ABD \leftrightarrow \triangle ACE$	i.
(i) $\overline{AB} \cong \overline{AC}$	(a) Given
(ii) $\angle A \cong \angle A$	(b) Common
(iii) $\overline{AD} \cong \overline{AE}$	(c) Halves of congruent sides are $\cong$
2. $\therefore \triangle ABD \cong \triangle ACE$	ii. S.A.S. $\cong$ S.A.S.
3. $\therefore \overline{BD} \cong \overline{CE}$	iii. By the congruence of triangles
Q.E.D.	

7. From a point not on the line, one and only one perpendicular can be drawn to the line.

8. Consider the figure given here and discuss the following questions/statements with your class-mates and try to find out the answers of the questions asked here.

- What relationship do you suggest between the pair of angles 1, 2 and 3, 4?
- Are $\angle 1$ and $\angle 4$ congruent? Are $\angle 3$ and $\angle 2$ congruent? If not, why? Under what Conditions they may be congruent?
- If $\angle 3$ and $\angle 2$ are congruent, what will happen to the two converging lines?

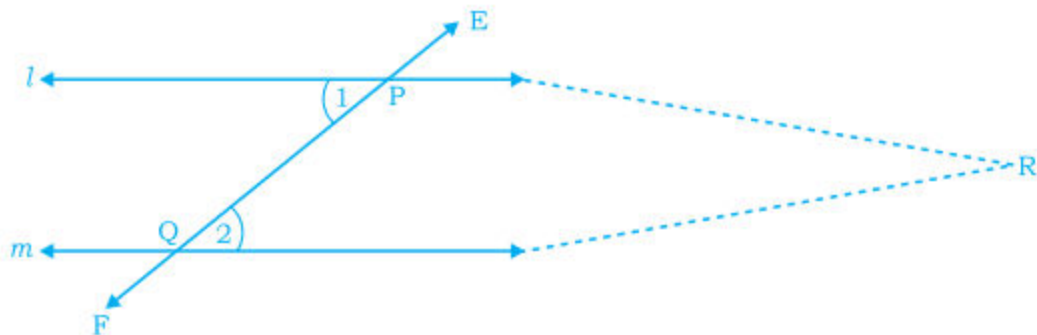


- (iv). $\angle 1, \angle 2$ and $\angle 3, \angle 4$ are supplementary angles. Do you agree? Give reasons. Are $\angle 1, \angle 3$ and $\angle 2, \angle 4$ also supplementary angles. If not, suggest a condition when these may be supplementary.
- (v). Find the condition when the pair of interior angles on the same side of a transversal are supplementary.
- (vi). If the converging lines are produced, they will meet at some point. What new figure will be developed? Out of four interior angles of converging lines which two of them will become the exterior angles of the triangle? Which exterior angle is greater than $m\angle 2$, and which exterior angle is greater than $m\angle 4$?

We will get answers to some of these questions after studying Theorems 7 and 8, so revisit these questions after we have learnt these theorems.

THEOREM 7

If a transversal intersects two coplanar lines such that the pair of alternate angles are congruent, then the lines are parallel.



Given: l and m are two coplanar lines and transversal $\overleftrightarrow{EF}$ intersects them at P and Q respectively, such that $\angle 1 \cong \angle 2$ (Pair of Alternate angles)

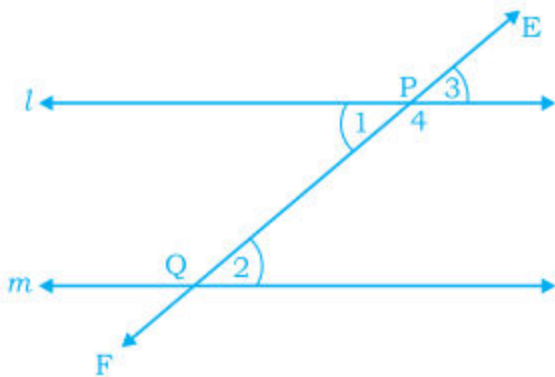
To Prove: $l \parallel m$

Proof: If l and m are parallel, then being coplanar they will meet at some point (say R) and PQR will be a triangle.

Statements	Reasons
1. In $\triangle PQR$, $\angle 1$ is exterior angle, and $\angle 2$ is interior opposite angle	i. By definition
2. $\therefore m\angle 1 > m\angle 2$	ii. By Theorem 6
3. But $m\angle 1 = m\angle 2$	iii. Given
4. Statements 2 and 3 cannot be true simultaneously	iv. Trichotomy Prop. of Order Rel.
5. So $m\angle 1 = m\angle 2$ is true and l, m do not meet.	v. Our supposition is wrong.
6. or $l \parallel m$	vi. They do not intersect
Q.E.D.	

Cor. 1. If a transversal intersects two coplanar lines such that the pair of corresponding angles are congruent then the lines are parallel.

Given: l and m are two coplanar lines and transversal $\overleftrightarrow{EF}$ intersects them at points P and Q respectively, such that $\angle 3 \cong \angle 2$ (Pair of Corresponding angles).



To Prove $l \parallel m$

Proof:

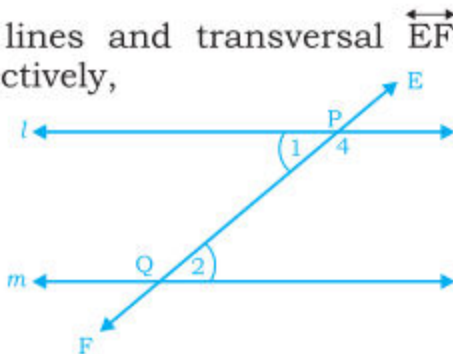
Statements	Reasons
1. $m\angle 1 = m\angle 3$	i. By Theorem 3
2. But $m\angle 3 = m\angle 2$	ii. Given
3. $\therefore m\angle 1 = m\angle 2$	iii. Transitive Prop. of Eq.
4. But these are alternate $\angle$ s, $\therefore l \parallel m$	iv. By Theorem 7
Q.E.D.	

Cor. 2. If a transversal intersects two coplanar lines such that the interior angles on the same side of the transversal are supplementary, then the lines are parallel.

Given: l and m are two coplanar lines and transversal $\overleftrightarrow{EF}$ intersects them at points P and Q respectively,

such that $m\angle 2 + m\angle 4 = 180^\circ$ (Pair of supplementary angles)

To Prove: $l \parallel m$



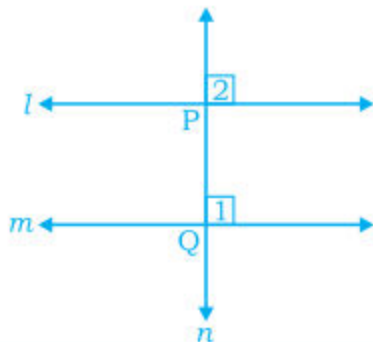
Proof:

Statements	Reasons
1. $\angle 1$ is the supplement of $\angle 4$	i. By Theorem 1
2. But $\angle 2$ is the supplement of $\angle 4$	ii. Given
3. $\therefore m\angle 1 = m\angle 2$	ii. Supplements of same angle are $\cong$
4. But these are alternate $\angle$ s, $\therefore l \parallel m$	iv. By Theorem 7
Q.E.D.	

Cor. 3. In a plane, if a line is perpendicular to each of two lines, then the two lines are parallel.

Give: l and m are two coplanar lines and transversal n intersects them at points P and Q respectively, such that $m\angle 1 = 90^\circ = m\angle 2$

To Prove: $l \parallel m$



Proof:

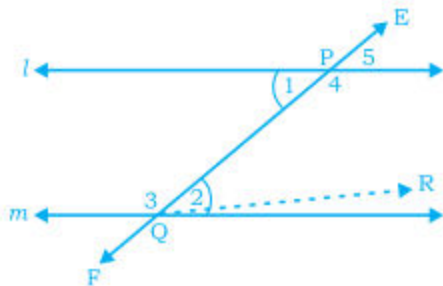
Statements	Reasons
1. $m\angle 1 = m\angle 2$	i. Each is a right angle (Given)
2. But these are Corresponding $\angle$ s, $\therefore l \parallel m$	ii. By Theorem 7 Corresponding $\angle$ s
Q.E.D.	

THEOREM 8

If a transversal intersects two parallel lines, the alternate angles so formed are congruent.

Given: l and m are two parallel

lines and transversal $\overleftrightarrow{EF}$ intersects them at points P and Q respectively.



To Prove : (i) $\angle 1 \cong \angle 2$ (ii) $\angle 3 \cong \angle 4$

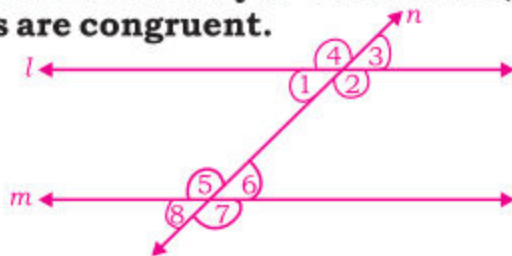
Proof: Suppose $\angle 1$ is not congruent to $\angle 2$, but $\angle 1 \cong \angle PQR$, where $\overrightarrow{QR}$ does not lie on m .

Statements	Reasons
1. Since $\angle 1 \cong \angle PQR$	i. By supposition
2. $\therefore l \parallel QR$	ii. By Theorem 7
3. But $l \parallel m$	iii. Given
4. $\therefore l$ is parallel to two lines m and QR which is impossible	iv. By Playfair's Axiom
5. $\therefore \angle 1 \cong \angle 2$	v. Our supposition leads to absurd result
6. Similarly, $\angle 3 \cong \angle 4$	vi. By the above process

Q.E.D

Cor. 1. If two parallel lines are intersected by a transversal, each pair of corresponding angles are congruent.

Given: $l \parallel m$ and transversal n intersects them forming eight angles as shown in the figure.



To Prove : $\angle 1 \cong \angle 8, \angle 2 \cong \angle 7, \angle 3 \cong \angle 6, \angle 4 \cong \angle 5$

(Pairs of Corresponding angles)

Proof:

Statements	Reasons
1. $l \parallel m$	i. Given
2. $\therefore \angle 1 \cong \angle 6$	ii. Alt. $\angle$ s (Theorem 8)
3. But $\angle 1 \cong \angle 3$	iii. Vertical Angles
4. $\therefore \angle 6 \cong \angle 3$	iv. Transitive Prop. of Eq.
5. Similarly, $\angle 1 \cong \angle 8, \angle 7 \cong \angle 2,$ $\angle 5 \cong \angle 4$	v. By the above process
Q.E.D.	

Cor. 2. If two parallel lines are intersected by a transversal, the interior angles on the same side of the transversal are supplementary.

Given: $l \parallel m$ and transversal n intersects them forming eight angles as shown in the figure. (See the figure of Cor. 1)

To Prove: $m\angle 1 + m\angle 5 = 180^\circ, m\angle 2 + m\angle 6 = 180^\circ$

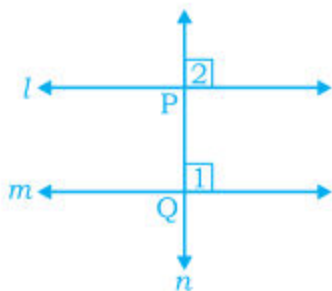
Proof:

Statements	Reasons
1. $l \parallel m$	i. Given
2. $\therefore m\angle 6 = m\angle 1$	ii. Alt. $\angle$ s (Theorem 8)
3. or $m\angle 6 + m\angle 2 = m\angle 1 + m\angle 2$	iii. Adding $m\angle 2$ on both sides
4. $\therefore m\angle 6 + m\angle 2 = 180^\circ$	iv. $m\angle 1 + m\angle 2 = 180^\circ$ (Th. 1)
5. Similarly $m\angle 1 + m\angle 5 = 180^\circ$	v. By the above process
Q.E.D.	

Cor. 3. In a plane, if a line is perpendicular to one of the two parallel lines, it is perpendicular to the other line also.

Given: $l \parallel m$ and n is perpendicular on m

To Prove: n is also perpendicular on l



Proof:

Statements	Reasons
1. $l \parallel m$	i. Given
2. $\therefore \angle 2 \cong \angle 1$ implies $\angle 2$ is a rt. $\angle$	ii. By Th. 8, Cor. 1 (Corresponding angles)
3. $\therefore n$ is also perpendicular on l	iii. $\angle 2$ is a rt. angle as proved

Q.E.D.

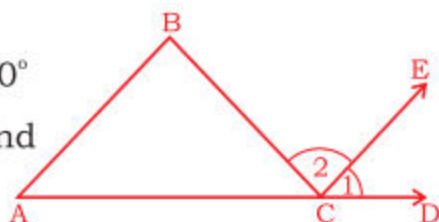
THEOREM 9

The sum of measures of the three angles of a triangle is 180° .

Given: $\triangle ABC$

To Prove : $m\angle A + m\angle B + m\angle C = 180^\circ$

Construction: Produce $\overrightarrow{AC}$ to point D and
Draw $\overrightarrow{CE}$ parallel to $\overrightarrow{AB}$.



Proof:

Statements	Reasons
1. $\overline{AB}$ is parallel to $\overrightarrow{CE}$, $\overrightarrow{AD}$ is transversal	i. Construction
2. $m\angle A = m\angle 1$	ii. Corresponding angles
3. $\overline{AB}$ is parallel to $\overrightarrow{CE}$, $\overline{BC}$ is transversal	iii. Construction
4. $m\angle B = m\angle 2$	iv. Alternate angles
5. $m\angle A + m\angle B = m\angle 1 + m\angle 2 = m\angle BCD$	v. Adding (2) and (4), Angle Addition Postulate
6. Adding $m\angle ACB$ to both sides $m\angle A + m\angle B + m\angle ACB = m\angle BCD + m\angle ACB$	vi. Addition Prop. of Eqs.
7. or $m\angle A + m\angle B + m\angle C = 180^\circ$	vii. $m\angle BCD + m\angle ACB = 180^\circ$

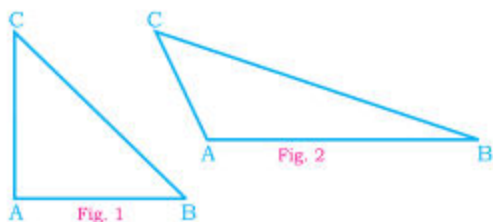
Q.E.D.

Cor. 1. In a triangle there can be only one right angle or one obtuse angle.

Given: $\triangle ABC$, $m\angle A = 90^\circ$

Or $m\angle A > 90^\circ$

To Prove : $\triangle ABC$ has only one
Right angle or only
One Obtuse angle.



Proof:

Statements	Reasons
1. In Fig. 1, $m \angle A = 90^\circ$	i. Given
2. $\therefore m \angle B + m \angle C = 180^\circ - 90^\circ = 90^\circ$ i.e., $\angle B$ and $\angle C$ are complementary $\angle$ s	ii. By Theorem 9 By Def. of Compl. Angles
3. $\therefore$ Each of $\angle B$ and $\angle C$ is less than a rt. $\angle$ or $\triangle ABC$ has only one right angle	iii. Their sum is 90° iv. Given
4. In Fig. 2, $m \angle A > 90^\circ$	v. Sum of measures of three angles is 180°
5. $\therefore m \angle B + m \angle C < 90^\circ$	vi. Sum of measures of these two angles is less than 90°
6. $\therefore$ Each of $\angle B$ and $\angle C$ is less than a rt. $\angle$ or $\triangle ABC$ has only one obtuse angle.	
Q.E.D.	

Cor. 2. Every triangle has at least two acute angles.

Given: $\triangle ABC$ (Refer the figures given above)

To Prove That: $\triangle ABC$ has at least two acute angles

Proof:

Statements	Reasons
1. If $\triangle ABC$ has only one Right or only one Obtuse angle, then it has two acute angles	i. By Theorem 9 Cor. 1
2. If $\triangle ABC$ has no Right or no Obtuse angle, then it has all the three angles acute	ii. A triangle has 3 angles
3. Hence, every triangle has at least two acute $\angle$ s.	iii. As proved above
Q.E.D.	

Cor. 3. In a right-angled triangle, acute angles are complementary. [For its solution, follow the first two steps of the proof of Cor. 1]

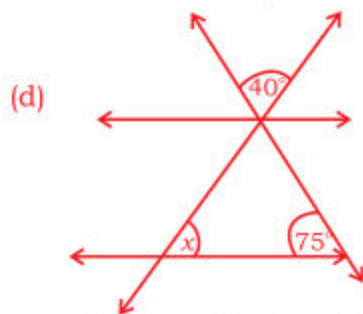
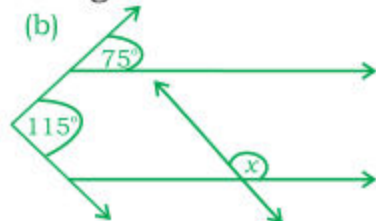
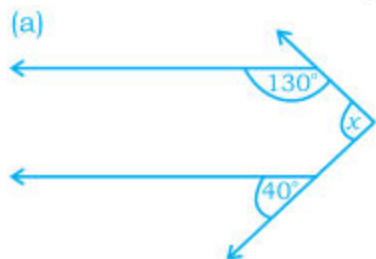
Cor. 4. From a point not on the line, one and only one perpendicular can be drawn to a given line. [Proved in Th. 6, Cor. 2]

Cor. 5. The measure of an exterior angle of a triangle is equal to the sum of the measures of non-adjacent interior angles of the triangle. [For solution, draw the figure of Th. 9, write 'Given' etc yourself, then

Cor. 6. If two angles of a triangle are congruent to two angles of another triangle, then its third angle will be congruent to the third angle of the other triangle [Left as an exercise for the students]

EXERCISE 10.5

1. In each of the figures given below, find the angle marked x . Numbers indicate the degree measure of the angle.



[**Hint:** From (a) to (d) through the vertex of angle marked x (or 135°) draw a line parallel to both the lines]

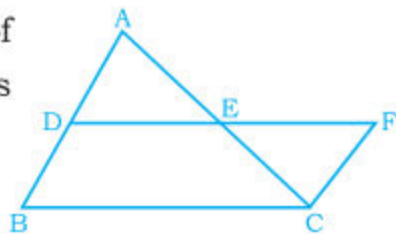
2. If two line segments $\overline{AB}$ and $\overline{CD}$ bisect one another at point O , prove that $\overline{AC}$ is parallel to $\overline{BD}$ and $\overline{AD}$ is parallel to $\overline{BC}$. [This problem has previously been solved. After proving the two triangles congruent use Theorem 7 to prove the lines to be parallel using alternate angles congruent.]

3. Points D and E are the mid-points of the sides $\overline{AB}$ and $\overline{AC}$ of a $\triangle ABC$. If $\overline{DE}$ is produced to point F so that $\overline{EF} \cong \overline{DE}$, prove that $\overline{CF} \parallel \overline{AB}$ and $\overline{DE} \parallel \overline{BC}$.

Given: Points D and E are the mid-points of the sides $\overline{AB}$ and $\overline{AC}$ of $\triangle ABC$. $\overline{DE}$ is produced to point F so that $\overline{EF} \cong \overline{DE}$

To Prove That:

$\overline{CF} \parallel \overline{AB}$ and $\overline{DE} \parallel \overline{BC}$.



Proof:

Statements	Reasons
1. In $\triangle ADE \leftrightarrow \triangle CFE$	i. 1-1 correspondence between two $\triangle$ s.
(A) $\overline{AE} \cong \overline{CE}$	(a) Given (E is the mid-point)
(B) $\angle AED \cong \angle CEF$	(b) Vertical angles
(C) $\overline{DE} \cong \overline{EF}$	(c) Given
2. $\therefore \triangle ADE \cong \triangle CFE$	ii. S.A.S. $\cong$ S.A.S.
3. $\therefore \angle A \cong \angle ECF$	iii. By the congruence of triangles
and $\overline{CF} \cong \overline{AD} \cong \overline{BD}$	iv. By Theorem 7
4. But $\angle A$ and $\angle ECF$ are alt. $\angle$ s	
$\therefore \overline{CF} \parallel \overline{AB}$ or $\overline{CF} \parallel \overline{BD}$	v. Proved above
5. $\overline{CF} \cong \overline{BD}$ and $\overline{CF} \parallel \overline{BD}$	vi. One pair of opposite sides of a quadrilateral is parallel and congruent, it is a parallelogram
6. $\therefore$ BCFD is a parallelogram	vii. Opposite sides of parallelogram are parallel
7. $\therefore \overline{DF} \parallel \overline{BC}$ or $\overline{DE} \parallel \overline{BC}$	viii. As proved above
8. or $\overline{CF} \parallel \overline{AB}$ and $\overline{DE} \parallel \overline{BC}$	

Q.E.D

4. Prove that opposite angles of a prallelogram are congruent.

[Hint: Use Cor. 2 of Theorem 8, i.e., $m\angle A + m\angle B = 180^\circ$ etc.]

5. How many degrees are there in an angle of an equilateral triangle?

6. Find measure of an angle which is equal to the sum of measures of other two angles of a triangle.

[Hint: By Theorem 9, Cor. 3, the sum of other two angles is 90° which is the same as the angle of a right angled triangle, so measure of the angle is 90° .]

7. The interior angles of a quadrilateral are together equal in measure to four right angles.

[Hint: Draw any one diagonal. Use Theorem 9 to prove that $2 \text{ rt } \angle + 2 \text{ rt } \angle = 4 \text{ rt } \angle$ s].

REVIEW EXERCISE 10

1. Demonstrative Geometry is a branch of Mathematics. What does it do?
2. **Write what you know about the following terms:**
(a) Undefined Terms (b) Assumptions or Fundamental Agreements
(c) Propositions or Theorems (d) Postulates
(e) Axioms (e) Corollaries
3. Write down the four Basics of Reasoning.
4. (a) State Theorem 1 and prove it.
(b) The sum of measures of all the angles at a point in a given straight line on the same side of it is equal to two right angles. Prove.
5. (a) Prove that if two lines intersect, then the opposite vertical angles are congruent.
(b) An equilateral triangle is also equiangular, prove.
6. (a) From a point not on the line, one and only one perpendicular can be drawn to a given line.
(b) If a transversal intersects two parallel lines, the alternate angles so formed are congruent.
7. (a) The measure of an exterior angle of a triangle is equal to the sum of the measures of non-adjacent interior angles of the triangle.
(b) State Theorem 5 and prove it.
8. **(a) Write 'True/False' against each statement:**
(i) If two lines intersect, then vertical adjacent angles are congruent.
(ii) Through a point not on a line, one and only one line can be drawn parallel to the line.
(iii) Sum of the measures of three angles of a triangle may be equal to 180° .
(iv) If a ray is drawn through a point of a given line and forms two adjacent angles, then these angles are complementary to one another.
(v) In a right angled triangle the acute angles are complementary.

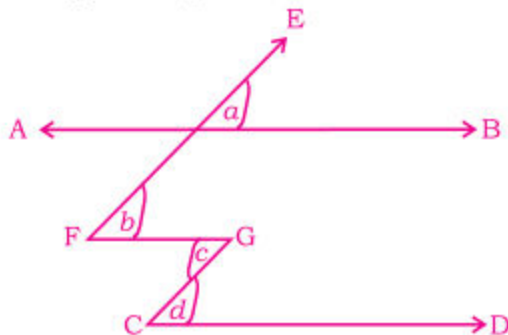
(B) Fill in the blanks with appropriate words:

- (i) Three non-collinear points determine a _____.
- (ii) Two distinct points determine a _____.
- (iii) One and only one _____ can be drawn from a point not on the line.
- (iv) If a line cuts two parallel lines, then each pair of _____ angles and each pair of _____ angles are congruent.
- (v) Measure of an exterior angle of a triangle is equal to the _____ of the measures of the non-adjacent Interior angles of the triangle.

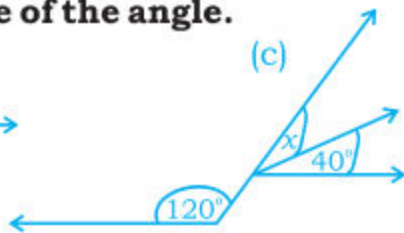
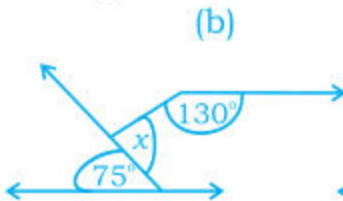
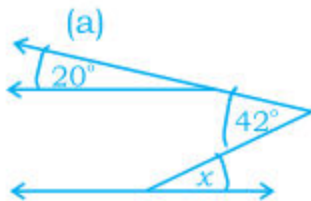
9. (i) In the figure on the right,

$$m\angle a = m\angle b = m\angle c = m\angle d$$

Write down four pairs of parallel lines.



(ii) In each of the figures given below, find the angle marked x . Numbers indicate the degree measure of the angle.



SUMMARY

Demonstrative Geometry is a branch of Mathematics in which statements concerning geometrical figures are proved through logical reasoning.

- Reasoning is the process of thinking about something in a logical way in order to form a conclusion or judgment.
- A reasoning in which we generalize specific observations into universal statements is called **Inductive Reasoning**.

- In **Deductive Reasoning** we deduce particular results from general results.
- The whole framework of Deductive Reasoning is based on the four elements known as **Basics of Reasoning** which are:
 1. Undefined terms
 2. Defined Terms
 3. Assumptions or Fundamental Agreements
 4. Propositions
- Fundamental Agreements (or Assumptions) are of two kinds.
- Axioms are those Fundamental Agreements which are related to numbers.
- Postulates are those Fundamental Agreements which are related to geometrical figures.
- **Proposition** is a statement proposed for discussion. Propositions are either **Theorems** or **Problems** (also called **Riders**)
- A proposition (Theorem or Rider) consists of two parts:
 1. **Hypothesis** and
 2. **Conclusion**
- A **corollary** is a fact which readily follows from a proved theorem.
- One theorem is said to be the converse of another theorem when the **hypothesis** of one is the **conclusion** of the other.
- **Enunciation** is the general statement of a geometrical truth which we are going to prove.
- **Given** or **Data** is the particular statement of the Hypothesis according to the figure which we have drawn.
- **To prove that** is the specific statement about the conclusion which we are going to prove.
- **Construction:** In this part, we add something to the given figure (by drawing a line, bisector etc) which may help in proving a theorem.
- **Proof:** In this last part of proving a theorem, the proposition is proved logically, putting statements in one column and its reason opposite to it in another column. The prescribed nine theorems follow this procedure.